Evolution of O.S

Lecture 1

Operating Systems

Simple Processing Systems

Serial processing systems were introduced in the **late 1940s** and were the dominant form of computer operation throughout the **1950s**. They represent the first generation of computing, where machines were operated without a formal operating system, requiring users to interact directly with the hardware.

Key Characteristics and Usage

•**No Operating System ("Bare Machine"):** There was no OS to manage resources; the user had exclusive,
•direct control of the machine for a scheduled amount of time.
•**Sequential Processing:** Jobs were executed one by one, in a strict serial (FIFO - First In, First Out) manner.
•**Input/Output Methods:** Programs were loaded using punched cards, paper tapes, and magnetic tapes.
•**Setup Time:** Users had to load compilers, linkers, and the program manually. If a program failed,
•the user had to restart the entire process.
•**"Sign-up" Scheduling:** Because machines were scarce, users signed up for dedicated time slots,
•leading to wasted time if a job finished early or encountered errors.

Simple batch systems

- Simple batch systems are early operating systems where users submit jobs (on punch cards) to an operator, who groups similar jobs into batches for sequential, non-interactive execution. They reduce setup time but create11 idle CPU time during slow I/O operations. A resident "batch monitor" handles automatic job sequencing.
- Examples: Early IBM Systems Like the FORTRAN IBSYS 7096.
- Simple batch systems were designed to maximize processing speed for large, repetitive, non-interactive tasks.

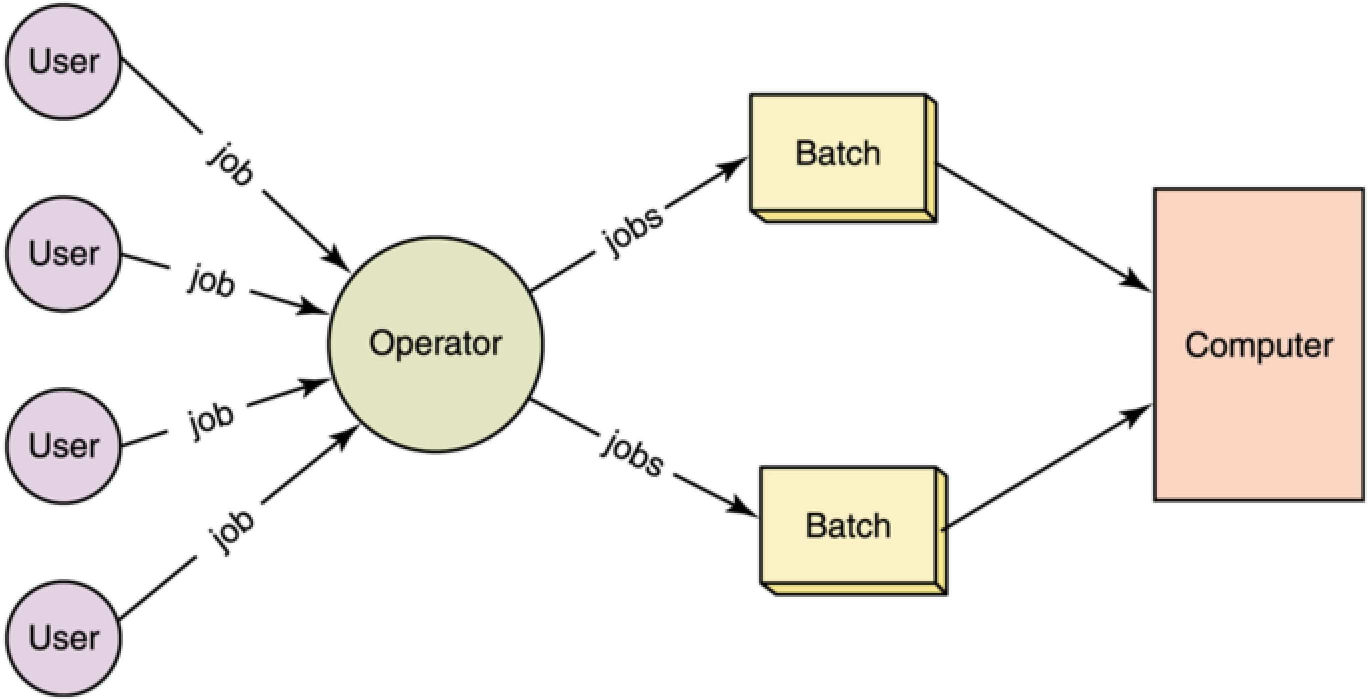
Key characteristics:

- **No Direct Interaction:** Users do not interact with the computer; jobs are submitted to an operator.
- **Batching Jobs:** Similar jobs (e.g., all FORTRAN jobs) are grouped together to speed up processing.
- **Sequential Execution:** Jobs are processed one after another.
- **Batch Monitor:** A primitive monitor program stays in memory to load the next job automatically.
- **Turnaround Time:** Output may take hours or days, as it is not real-time.

Workflow:

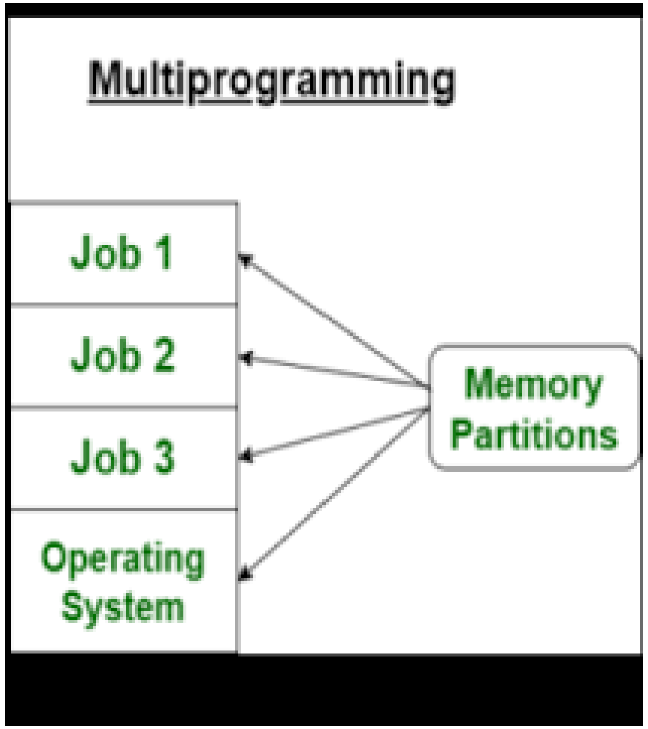
- **Job Submission:** Users prepare programs/data on punch cards and give them to an operator.
- **Batch Creation:** The operator groups similar jobs.
- **Execution:** The monitor reads jobs, executes them, and prints output.
- **Result Delivery:** The operator returns results (output, memory dump) to the user.

Simple batch systems



Multiprogrammed batch Systems

- Multiprogrammed batch systems enhance CPU efficiency by loading multiple jobs into memory simultaneously, switching between them to eliminate idle time, particularly during I/O operations.
- Unlike simple batch systems, they allow the operating system to execute a new job when the current one waits, maximizing throughput for repetitive tasks.



Time-sharing systems

- Time-sharing systems allow multiple users to interact with a single computer simultaneously by dividing CPU time into rapid, small intervals called time slices (10-100 ms). By quickly switching between users, the operating system gives each user the illusion of exclusive access to the machine. It is a logical extension of multiprogramming designed to improve responsiveness and maximize CPU efficiency.
- **Time Slicing (Quantum):** Each user/task gets a small, fixed amount of CPU time.
- **Interactive Computing**: Users can interact with their programs in real-time, receiving immediate feedback.
- **CPU Scheduling**: The OS rapidly switches between users, reducing CPU idle time.
- **Independent Sessions**: Although resources are shared, each user operates in an independent session.